

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA

Expected May 2028

Expected May 2027

Relevant Coursework: Introduction to Algorithms, Introduction to Computer Systems, Automata, Computability & Complexity, Programming Languages & Techniques, Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Made the in-game AI companion fight only what it should** — its targeting treated anything flagged hostile as fair game, dragging children into boss fights they could not survive; diagnosed it and shipped a fix that can only ever remove a bad target, never add one or weaken defence against real hostiles
- **Packaged the in-game AI agent as an SDK outside studios can build on** — versioned, sandboxed per player session, covered by **1,301 tests** — proven by an external builder shipping a Minecraft mode with no repository access, loading their own content at runtime and reasoning about items it had never seen
- **Extended the agent's perception to modded content** — it learns a mode's custom blocks and items at runtime, so it names what a child is holding instead of reading back a raw registry ID it has never seen
- Delivered an app-wide **React Native** redesign from Figma to production — shipped the reorganized screens first so users kept their bearings, then rolled a **33-section** design system onto every screen

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Improved UI/UX and core gameplay stability across **50+ live Unity titles** in a **420M+ download** portfolio
- Shipped fixes for **100+ bugs** raised by QA and player reports, correcting Unity prefabs and C# scripts in production
- Kept all 50+ titles live on the app stores by upgrading SDKs and platform modules as requirements changed

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a team of **15+** designers and developers shipping new products for real clients, from first concept to launch
- Defined the end-to-end user flows and reusable interface systems the team shipped across multiple client deliverables

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **A Minecraft-style 3D game engine written from scratch in C++ and OpenGL** (team of 3) — I owned the world generation: **7 procedural biomes** blended from layered noise, and caves generated as you explore
- **Owned the rendering pipeline that makes the world look real** — **13 GLSL shaders** for shadows, reflective water, ambient shading and a day-night cycle, plus post-process overlays for underwater and lava

Passenger | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- **An animated short film built in Unreal Engine 5** — I wrote **2 custom shaders** that make the ray-traced render read as hand-painted rather than computer-generated, rendered out through the Movie Render Pipeline
- **Rigged and animated the character in Maya**, driven inside Unreal through an animation blueprint

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **A 3D modeling tool in the spirit of Maya, in C++ and Qt** — edit any model's geometry directly, with a half-edge structure tracking how faces, edges and vertices connect, each selectable from a clickable Qt list
- **Smooths rough models into refined surfaces** with Catmull–Clark subdivision, the algorithm film and game studios use

TECHNICAL SKILLS

Game Engines: Unity, Unreal Engine 5, C#, C++, Animation Blueprints, prefab & gameplay systems

Graphics & 3D: OpenGL, GLSL, HLSL, Maya, Blender, Substance Painter, Movie Render Pipeline, Lumen

Languages: C++, C#, Python, TypeScript, JavaScript, Java, Kotlin, SQL, HLSL/GLSL

Frameworks, Infra & AI: React Three Fiber, React Native (Expo), Minecraft modding, Git, Docker, Claude Code, Codex, MCP

Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing